

OOPS EXPERIMENT-1

EXPERIMENT – 1

TITLE: Introduction to Java Environment

1. Explore and understand the role of JDK, JRE and JVM.
2. Install latest available JDK and verify the Java Environment.
3. Create a Sample Hello World Program using simple text editor (e.g. Notepad) and show the steps to compile and execute the program using command prompt.
4. Display your name and complete address in different lines.

ANSWER-1

When we write and run Java programs, three key components help make it possible for us to do so :- **JDK, JRE, and JVM.**

1. JVM(Java virtual machine) :- The Heart of java

We can think of **JVM** as a magic box that allows Java programs to run on any computer, no matter which operating system (Windows, Mac, Linux) you use. It converts Java code into machine language that your computer understands.

Java is "Write Once, Run Anywhere" because JVM makes sure the same program runs on different computers without changes.

2. JRE (Java Runtime Environment) – Helps Run Java Programs

We can think of JRE as a kit that contains everything needed to run Java programs . It includes the JVM and necessary libraries to run Java applications. Without JRE, you cannot run Java programs. If you just want to run Java applications and not to develop them, you only need the JRE.

3. JDK (Java Development Kit) – Helps Create Java Programs

JDK Is a toolbox for java developers. It includes **JRE + JVM + Development Tools**. You need JDK if you want to write and compile Java programs. Without JDK, you cannot develop Java applications.

Component	Purpose	Includes
JVM	Runs Java programs	Part of JRE
JRE	Provides the environment to run Java applications	Includes JVM + libraries
JDK	Used to develop Java programs	Includes JRE + JVM + development tools

ANSWER -2

```
Command Prompt
Microsoft Windows [Version 10.0.26100.3190]
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C:\Users\Lenovo>java -version
java version "22.0.1" 2024-04-16
Java(TM) SE Runtime Environment (build 22.0.1+8-16)
Java HotSpot(TM) 64-Bit Server VM (build 22.0.1+8-16, mixed mode, sharing)

C:\Users\Lenovo>javac -version
javac 22.0.1

C:\Users\Lenovo>
```

ANSWER-3

```
Windows PowerShell
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PS C:\Users\Lenovo> cd desktop
PS C:\Users\Lenovo\desktop> javac HelloWorld.java
PS C:\Users\Lenovo\desktop> java HelloWorld
Hello, World!
PS C:\Users\Lenovo\desktop> |
```

ANSWER- 4

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Windows PowerShell
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PS C:\Users\Lenovo> cd desktop
PS C:\Users\Lenovo\desktop> javac MyDetails.java
PS C:\Users\Lenovo\desktop> java MyDetails
Priyambad Suman
123 Street Name
City, State, ZIP Code
PS C:\Users\Lenovo\desktop> |
```